21/08/2024

LSM

$$Y_i = \beta_0 + \beta_1 X_i^{(1)} + \beta_2 X_i^{(2)} + \varepsilon_i \qquad i = 1, \ldots, n$$

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}$$

Solve the least squares estimation problem in two steps

$$Y_i = \beta_0 + \beta_1 X_i^{(1)} + \varepsilon_i$$

Then "augment" variable $X^{(2)}$ to the model.

If $\underline{X}^{(2)} = \begin{pmatrix} X_1^{(2)} \\ \vdots \\ X_n^{(2)} \end{pmatrix}$ is orthogonal to $\underline{1}_n$ and $\underline{X}^{(1)} = \begin{pmatrix} X_1^{(1)} \\ \vdots \\ X_n^{(1)} \end{pmatrix}$

then there will be no change in the least squares estimate for $\begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}$ whether we use $\mathcal{M}_1$ or $\mathcal{M}_2$ to fit the data

$$\mathbb{X} = [1_n : \underline{X}^{(1)} : \underline{X}^{(2)}] = [\mathbb{Z} : X^{(2)}]$$

From Normal Equ$^n$ $\hat{\beta} = (\mathbb{X}^T \mathbb{X})^{-1} \mathbb{X}^T Y = \begin{pmatrix} \mathbb{Z}^T \mathbb{Z} & 0 \\ 0 & X^{(2)T} X^{(2)} \end{pmatrix}^{-1} \begin{pmatrix} \mathbb{Z}^T Y \\ X^{(2)T} Y \end{pmatrix}$

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \begin{pmatrix} (\mathbb{Z}^T \mathbb{Z})^{-1} \mathbb{Z}^T Y \\ (X^{(2)T} X^{(2)})^{-1} X^{(2)T} Y \end{pmatrix}$$

$$\frac{\|(I-P_{\mathbb{Z}})\underline{X}^{(2)}\|_2^2}{\|\underline{X}^{(2)}\|_2^2} \longrightarrow \text{small indicates "almost" belongs to } \mathcal{C}(\mathbb{X}).$$

$$Y = \mathbb{Z}\begin{pmatrix}\beta_0\\ \beta_1\end{pmatrix} + \beta_2 X^{(2)} + \varepsilon$$

$$(I-P_{\mathbb{Z}})Y = \beta_2 (I-P_{\mathbb{Z}})X^{(2)} + \tilde{\varepsilon} \qquad \mathcal{M}_3$$

Claim: The least squares estimate, say $\tilde{\beta}_2$, of $\beta_2$ from model $\mathcal{M}_3$, is the same as $\hat{\beta}_2$, the least squares estimate of $\beta_2$ from model $\mathcal{M}_2$ (full model), provided $(I-P_{\mathbb{Z}})\underline{X}^{(2)} \neq 0$

$$\Rightarrow \hat{\beta}_2 \overset{?}{=} \tilde{\beta}_2 = \frac{((I-P_{\mathbb{Z}})X^{(2)})^T((I-P_{\mathbb{Z}})Y)}{((I-P_{\mathbb{Z}})X^{(2)})^T((I-P_{\mathbb{Z}})X^{(2)})}$$

$$\tilde{\beta}_2 = \tilde{\tilde{\beta}}_2$$

$$= \frac{X^{(2)T}(I-P_{\mathbb{Z}})Y}{X^{(2)T}(I-P_{\mathbb{Z}})X^{(2)}}$$

l.s. estimate of $\beta_2$ from $Y = \beta_2(I-P_{\mathbb{Z}})X^{(2)} + \varepsilon$

## General version

$$Y = \mathbb{X}\beta + \varepsilon = \mathbb{Z}\gamma + \mathbb{W}\delta + \varepsilon \qquad \underset{p\times 1}{\beta} = \begin{pmatrix}\underline{\gamma}_{q\times 1}\\ \underline{\delta}_{r\times 1}\end{pmatrix}$$

$$\underset{n\times p}{\mathbb{X}} = [\underset{n\times q}{\mathbb{Z}} : \underset{n\times r}{\mathbb{W}}]$$

$\mathcal{M}_{red}: Y = \mathbb{Z}\underline{\gamma} + \varepsilon$

(when $\underline{\delta} = \underline{0}$)

<u>Claim</u>: The least squares estimate $\hat{\beta}$ from $\mathcal{M}_{full}$ can be expressed as $\hat{\beta} = \begin{pmatrix} \hat{\gamma} \\ \hat{\delta} \end{pmatrix}$ where

$$\hat{\gamma} = \left(\mathbb{Z}^T(I - P_{\mathbb{W}})\mathbb{Z}\right)^{-1}\mathbb{Z}^T(I - P_{\mathbb{W}})\underline{Y}$$

$$\hat{\delta} = \left(\mathbb{W}^T(I - P_{\mathbb{Z}})\mathbb{W}\right)^{-1}\mathbb{W}^T(I - P_{\mathbb{Z}})\underline{Y}$$

Furthermore, $\hat{\delta} = \tilde{\delta}$ where $\tilde{\delta}$ is the l.s. estimate of $\delta$ from the model $(I - P_{\mathbb{Z}})\underline{Y} = (I - P_{\mathbb{Z}})\mathbb{W}\delta + \tilde{\varepsilon}$

<u>Proof of Claim</u>:

Approach (1) $(\mathbb{X}^T\mathbb{X})^{-1} = \begin{pmatrix} \mathbb{Z}^T\mathbb{Z} & \mathbb{Z}^T\mathbb{W} \\ \mathbb{W}^T\mathbb{Z} & \mathbb{W}^T\mathbb{W} \end{pmatrix}^{-1}$

Use block matrix inversion formula with $P_{\mathbb{Z}} = \mathbb{Z}(\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T$

<u>Lemma</u>: $P_{\mathbb{X}} = P_{\mathbb{Z}} + P_{(I - P_{\mathbb{Z}})\mathbb{W}} = P_{\mathbb{Z}} + P_{\mathbb{W}|\mathbb{Z}}$

orthogonal projectors onto mutually orthogonal subspaces

Approach II (make use of the above lemma)

Fitted value from full model is $P_{\mathbb{X}} Y = \hat{Y} = \mathbb{X}\hat{\beta} = \mathbb{Z}\hat{\gamma} + \mathbb{W}\hat{\delta}$

By the lemma, where

$$P_{\mathbb{Z}} Y + P_{\mathbb{W}|\mathbb{Z}} Y = \mathbb{Z}\hat{\gamma} + \mathbb{W}\hat{\delta}$$

Premultiply by $(I - P_{\mathbb{Z}})$

$$(I-P_{\mathbb{Z}})P_{\mathbb{W}|\mathbb{Z}} Y = (I-P_{\mathbb{Z}})\mathbb{W}\hat{\delta}$$

Observe that

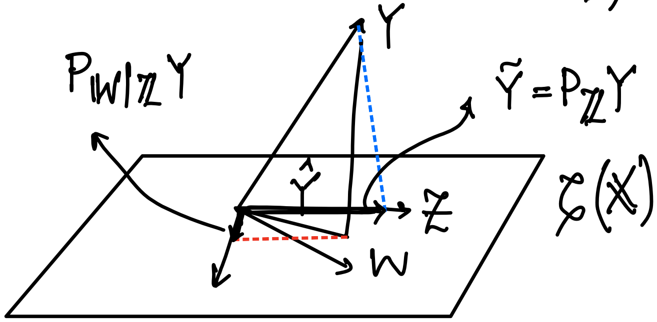
$$P_{\mathbb{W}|\mathbb{Z}} = (I-P_{\mathbb{Z}})\mathbb{W}\left(\mathbb{W}^T(I-P_{\mathbb{Z}})\mathbb{W}\right)^{-1}\mathbb{W}^T(I-P_{\mathbb{Z}})$$

Premultiply by $\mathbb{W}^T$

$$\mathbb{W}^T P_{\mathbb{W}|\mathbb{Z}} Y = \mathbb{W}^T(I-P_{\mathbb{Z}})\mathbb{W}\hat{\delta}$$

$$\Rightarrow \hat{\delta} = \left(\mathbb{W}^T(I-P_{\mathbb{Z}})\mathbb{W}\right)^{-1}\mathbb{W}^T(I-P_{\mathbb{Z}})\mathbb{W}\left(\mathbb{W}^T(I-P_{\mathbb{Z}})\mathbb{W}\right)^{-1}\mathbb{W}^T(I-P_{\mathbb{Z}})Y$$

$$= \left(\mathbb{W}^T(I-P_{\mathbb{Z}})\mathbb{W}\right)^{-1}\mathbb{W}^T(I-P_{\mathbb{Z}})Y \qquad [QED]$$



Proof of Lemma

$$P_{\mathbb{X}} = P_{\mathbb{Z}} + P_{(I-P_{\mathbb{Z}})W}$$

Show that $\underset{\sim}{u} \in \mathcal{C}(\mathbb{X})$

$$\Leftrightarrow \quad \underset{\sim}{u} = P_{\mathbb{Z}}\underset{\sim}{u} + P_{\mathbb{W}|\mathbb{Z}}\underset{\sim}{u} \text{ ——— } **$$

If $u = P_{\mathbb{Z}}u + P_{\mathbb{W}|\mathbb{Z}}u$

$$= P_{\mathbb{Z}}u + P_{(I-P_{\mathbb{Z}})W}u$$

$$= P_{\mathbb{Z}}u + (I-P_{\mathbb{Z}})W\left(W^T(I-P_{\mathbb{Z}})W\right)^{-1}W^T(I-P_{\mathbb{Z}})u$$

$$= \underbrace{P_{\mathbb{Z}}(\ )}_{} + \underbrace{W(\ \ )}_{\in \mathcal{C}(\mathbb{X})}$$

Conversely suppose $u \in \mathcal{C}(\mathbb{X})$, then

$$u = \mathbb{Z}v_1 + \mathbb{W}v_2$$

$$\Rightarrow (I-P_{\mathbb{Z}})u = (I-P_{\mathbb{Z}})\mathbb{W}v_2$$

$$u = \mathbb{Z}v_1 + \mathbb{W}v_2$$

$$= \mathbb{Z}v_1 + P_{\mathbb{Z}}\mathbb{W}v_2 + (I-P_{\mathbb{Z}})Wv_2$$

$$= P_{\mathbb{Z}}(\underbrace{\mathbb{Z}v_1 + \mathbb{W}v_2}_{u}) + (I-P_{\mathbb{Z}})Wv_2$$

We need to verify that

$$P_{\mathbb{W}|\mathbb{Z}}\, u = (I - P_{\mathbb{Z}})\mathbb{W} v_2$$

$$\text{LHS} = (I - P_{\mathbb{Z}})\mathbb{W}\left(\mathbb{W}^T(I-P_{\mathbb{Z}})\mathbb{W}\right)^{-1}\mathbb{W}^T(I-P_{\mathbb{Z}})(\mathbb{Z}v_1 + \mathbb{W}v_2)$$

$$= (I - P_{\mathbb{Z}})\mathbb{W} v_2$$

<u>Implication</u>:

$$Y = \mathbb{Z}\gamma + \mathbb{W}\delta + \varepsilon \quad (\mathcal{M}_{full})$$

$$Y = \mathbb{Z}\gamma + \varepsilon \quad (\mathcal{M}_{red})$$

Compare the SSE of the two models

$$SSE_{full} \le SSE_{red} \qquad \begin{pmatrix}\gamma \\ 0\end{pmatrix} = \beta\Big|_{\gamma=0}$$

$$\min_{\gamma,\delta} \|Y - (\mathbb{Z}\gamma + \mathbb{W}\delta)\|^2 \qquad \min_{\gamma} \|Y - \mathbb{Z}\gamma\|^2$$

$$SSE_{full} = Y^T(I - P_{\mathbb{X}})Y$$

$$SSE_{red} = Y^T(I - P_{\mathbb{Z}})Y$$

$$SSE_{red} - SSE_{full} = Y^T(P_{\mathbb{X}} - P_{\mathbb{Z}})Y$$

$$= Y^T P_{\mathbb{W}|\mathbb{Z}} Y$$

If $\dfrac{SSE_{red} - SSE_{full}}{SSE_{full}}$ is "small" then that indicates that the reduced model may be "adequate".

$$(H_0: \delta = 0 \text{ vs } H_A: \delta \neq 0)$$

A more general way to impose restriction on linear models is through specification of <u>linear constraints</u> on the regression parameter

$$\underset{n\times 1}{Y} = \underset{n\times p}{X}\underset{p\times 1}{\beta} + \underset{n\times 1}{\varepsilon}$$

restriction

$$\underset{m\times p}{L}^T \underset{p\times 1}{\beta} = \underset{m\times 1}{\theta_0}$$

(a set of $m \geq 1$ linear restriction)

Then arises in testing hypothesis of the kind

$$H_0 : L^T\beta = \theta_0 \quad \text{vs.} \quad H_A : L^T\beta \neq \theta_0$$

(general linear hypothesis)

<u>Result</u>: Suppose $\operatorname{rank}(X) = p$

$\operatorname{rank}(L) = m \leq p$ (no redundancy in the linear restrictions)

Let $\tilde{\beta} = \underset{\beta : L^T\beta = \theta_0}{\operatorname{argmin}} \|Y - X\beta\|_2^2$

$$\tilde{\beta} = \hat{\beta} - (X^TX)^{-1} L \left(L^T (X^TX)^{-1} L\right)^{-1} (L^T\hat{\beta} - \theta_0)$$

where $\hat{\beta}$ = unrestricted l.s. estimate

Proof by technique of Lagrange Multiplier.